Ch 6: Supply, Demand, and Government Policies

Price ceilings, price floors, and who really pays a tax

Onur Altındağ

EC 111 Principles of Microeconomics, Bentley University, Fall 2026

From markets to policy

Chapters 4–5

Markets set P and Q on their own

Elasticity: how far each one moves

Chapter 6

The government steps in:

- a price ceiling
- a price floor
- a tax

Massachusetts minimum wage: $15.00 an hour

Federal: $7.25, unchanged since 2009

Rent control: banned statewide by a 1994 ballot question, back in the debate

Question

Who gains, who loses, and does the policy do what it promises?

Monday: price controls · Wednesday: taxes and who bears them

Two kinds of price control

Price ceiling	Price floor
Legal maximum price	Legal minimum price
Rent control, gasoline 1973–79, price-gouging laws	Minimum wage, farm price supports

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Ceiling above the equilibrium price, floor below it: not binding, nothing changes

Ceiling below equilibrium, floor above it: binding, the market cannot clear

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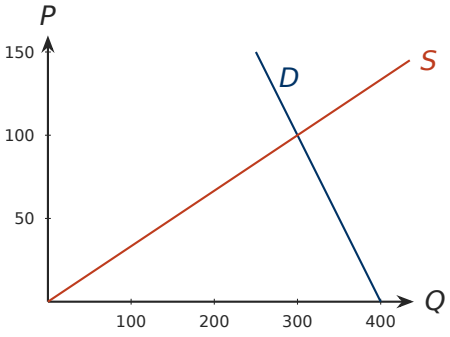
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Who asks for which: buyers lobby for ceilings, sellers for floors.

The market we will use

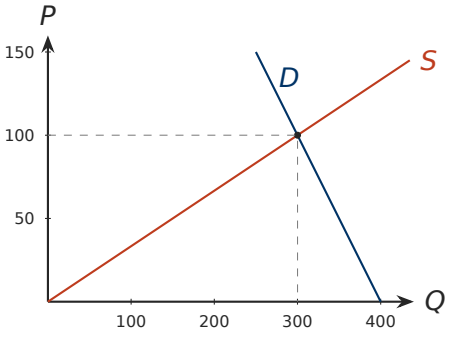


$$Q_d = 400 - P \qquad Q_s = 3P$$

Question

Find the equilibrium price and quantity.

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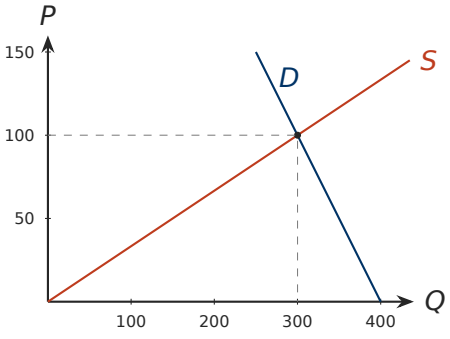
$$Q_d = 400 - P \qquad Q_s = 3P$$

Question

Find the equilibrium price and quantity.

$$400 - P = 3P \Rightarrow P^* = 100, \quad Q^* = 300$$

The market we will use



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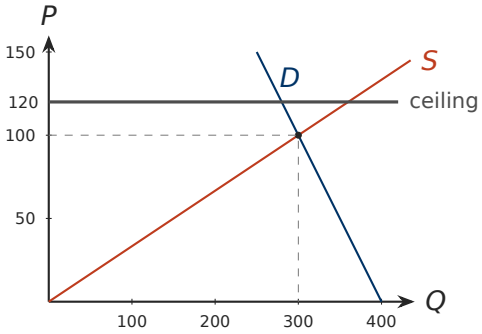
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Every policy today starts from here.

A ceiling at $120: nothing happens

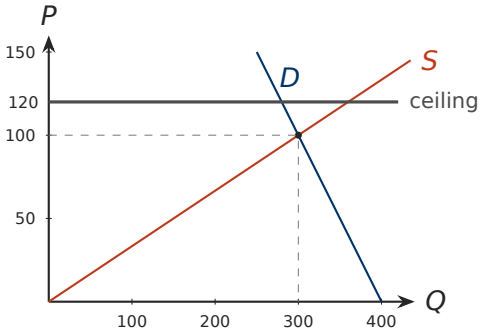


Legal maximum $120

Market price $100: already legal

$P = 100$, $Q = 300$, as before

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Legal maximum $120

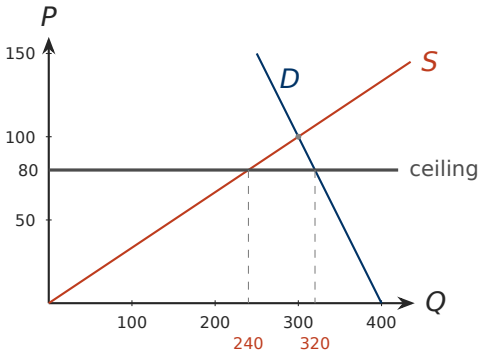
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Not binding

A ceiling above the equilibrium price has no effect

A ceiling at $80: a shortage

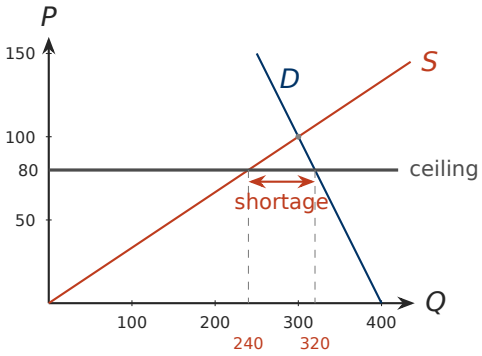


Ceiling \$80, below $P^* = 100$

$Q_s = 3 \times 80 = 240$

$Q_d = 400 - 80 = 320$

A ceiling at $80: a shortage



Ceiling \$80, below $P^* = 100$

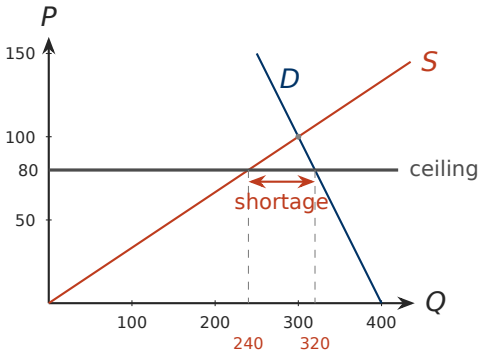
$Q_s = 3 \times 80 = 240$

$Q_d = 400 - 80 = 320$

Shortage $= 320 - 240 = 80$

Units sold: 240, not 300

A ceiling at $80: a shortage



Ceiling \$80, below $P^* = 100$

$Q_s = 3 \times 80 = 240$

$Q_d = 400 - 80 = 320$

Shortage $= 320 - 240 = 80$

Units sold: 240, not 300

Binding

The price cannot rise to clear the market. The shortage stays.

If not the price, then what decides who gets it?

320 buyers want it, 240 units exist

Something else does the rationing:

Lines — time wasted, 1973 and 1979 gas stations

Seller's choice — friends, relatives, "no pets, no kids"

Side payments — key money, black markets

Quality falls — the landlord stops fixing things

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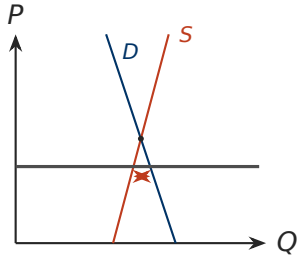
Buyers who get a unit pay less

80 buyers who wanted one at $80 get nothing

60 units that would have been traded at $100 are never produced

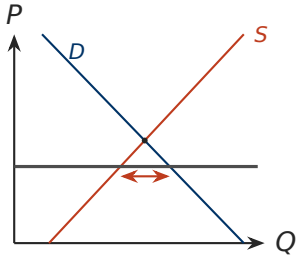
Rent control: short run vs. long run

Short run



steep curves: small shortage

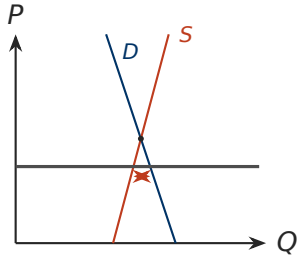
Long run



flat curves: large shortage

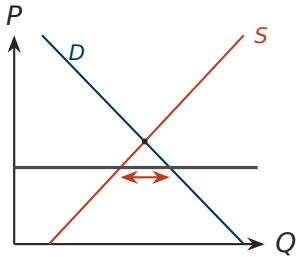
Rent control: short run vs. long run

Short run



steep curves: small shortage

Long run



flat curves: large shortage

Same ceiling. Over years landlords stop building and maintaining; more people want their own place. Both curves get more elastic, and the shortage grows.

Rent control: the evidence

Cambridge, MA

Rent control ended 1995 (1994 ballot)

Decontrolled buildings rose in value — and so did never-controlled buildings nearby

Autor, Palmer and Pathak, *JPE* 2014

San Francisco

1994: rent control extended to small older buildings

Covered tenants stayed longer

Landlords converted units to condos: rental supply in those buildings down about 15%, rents up city-wide

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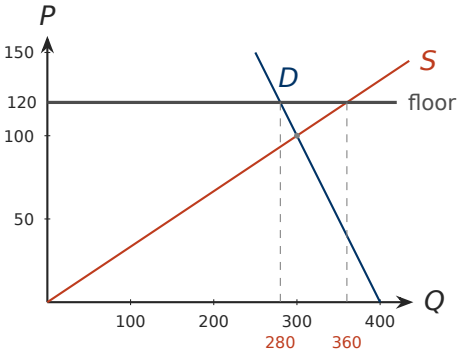
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Today's tenants gain; tomorrow's renters pay, through fewer units and higher rents elsewhere.

A price floor at $120: a surplus

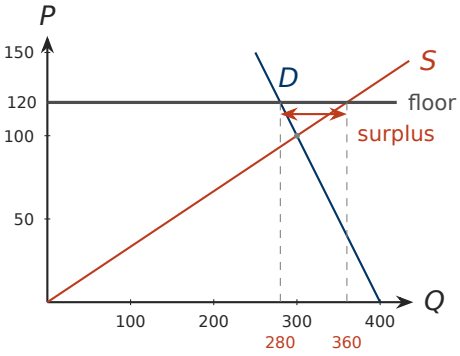


Floor \$120, above $P^* = 100$

$Q_d = 400 - 120 = 280$

$Q_s = 3 \times 120 = 360$

A price floor at $120: a surplus



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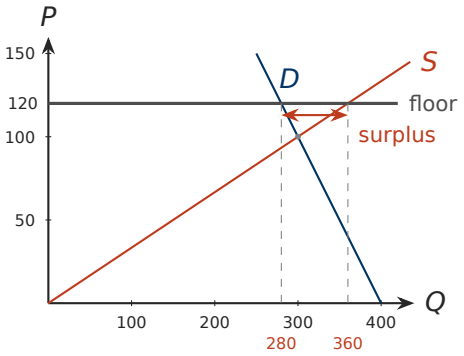
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Surplus $= 360 - 280 = 80$

Units sold: 280, not 300

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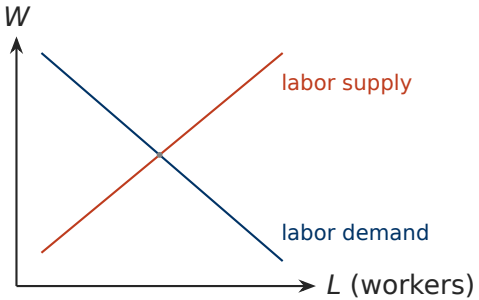
$Q_s = 3 \times 120 = 360$

Surplus $= 360 - 280 = 80$

Units sold: 280, not 300

A floor at \$80 would be not binding: \$100 is already above it.

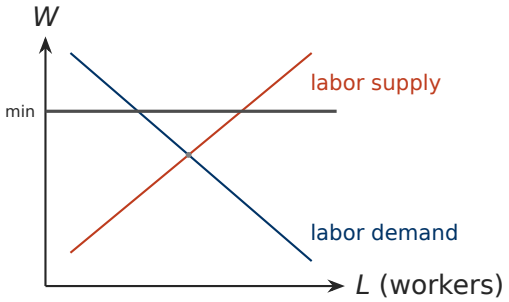
The minimum wage



Price of labor = the wage

Firms demand labor, workers supply it

The minimum wage

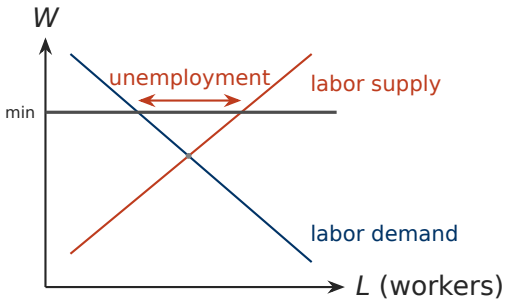


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Minimum above the market wage: a binding floor

The minimum wage



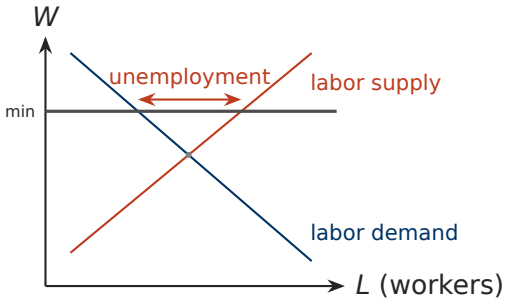
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More want jobs, firms want fewer: unemployment

The minimum wage



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Binds for young and low-skill workers

Jobs lost depend on the elasticity of labor demand

The minimum wage: what the numbers say

Two sides of the same floor

Workers who keep the job: higher pay

Workers who lose it or never get hired: nothing

Evidence

Card and Krueger (1994): NJ raised it, PA did not — no fall in fast-food jobs

Later work: small effects for modest rises

CBO (2019): a $15 federal minimum by 2025

Pay raised for 17 million

1.3 million out of poverty

Jobs lost: median estimate 1.3 million, range 0 to 3.7 million

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Question

Who wants it raised: a worker who has the job, or a 17-year-old looking for a first one?

Evaluating price controls

The goal is usually fair

Cheaper housing for the poor

A living wage

The tool is blunt

Prices ration and signal (Ch 4, Hayek)

Fix the price and you lose both

Shortage or surplus, and fewer trades

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Rent control $\rightarrow$ housing vouchers: the rent rises somewhat and more apartments are supplied; tenants pay less out of pocket

Minimum wage $\rightarrow$ Earned Income Tax Credit: the wage employers pay falls somewhat and more people work; take-home pay rises

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The cost moves to taxpayers, and the market still clears.

Try it

Question

Same market: $Q_d = 400 - P$, $Q_s = 3P$.

(a) A price ceiling at \$60. Binding? Shortage or surplus, and how big? How many units are sold?

(b) A price floor at \$110. Same questions.

(c) A price ceiling at \$150. Same questions.

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(a) Binding. $Q_s = 180$, $Q_d = 340$: shortage 160, 180 sold.

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(b) Binding. $Q_d = 290$, $Q_s = 330$: surplus 40, 290 sold.

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(c) Not binding: \$100 is legal. $P = 100$, $Q = 300$.

Taxes

Governments tax goods to raise revenue and to discourage use

Gasoline, cigarettes, alcohol, hotel rooms, Red Sox tickets

The law says who sends the check

The market decides who pays

Tax incidence

How the burden of a tax is shared between buyers and sellers

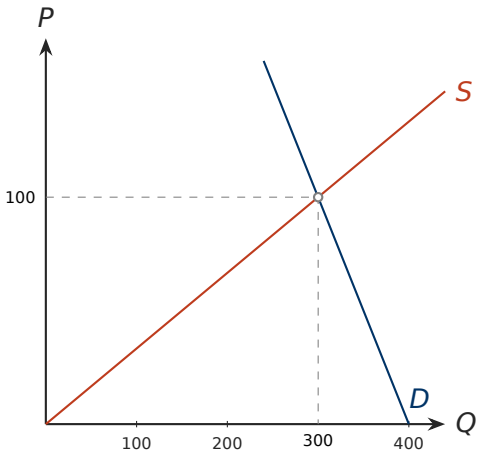
Three questions for any tax:

What happens to Q?

What price do buyers pay, what price do sellers keep?

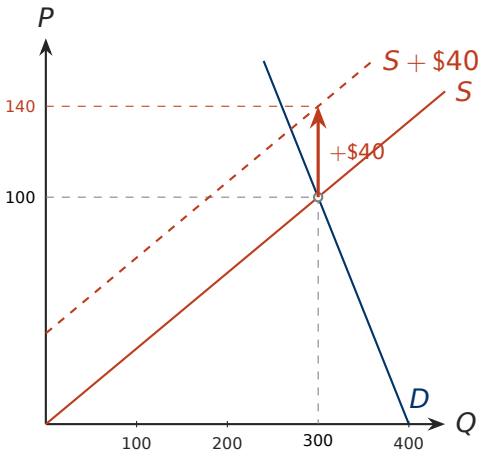
Does it matter who the law taxes?

A $40 tax on sellers



Sellers owe $40 on every unit sold

A $40 tax on sellers



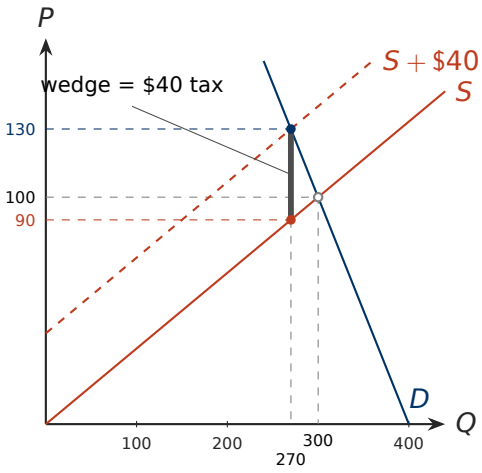
Sellers owe $40 on every unit sold

Before: 300 units at $100

Now they need $140 for them

S moves up $40: $Q_s = 3(P - 40)$

A $40 tax on sellers



Sellers owe $40 on every unit sold

Before: 300 units at $100

Now they need $140 for them

S moves up $40: $Q_s = 3(P - 40)$

At $140 buyers want only 260: P falls

$400 - P = 3P - 120 \Rightarrow P = 130$

Buyers pay **130**, sellers keep **90**,

$Q = 270$

Who paid the $40?

	Before	After	Change
Buyers pay	$100	$130	+30
Sellers keep	$100	$90	−10
Quantity	300	270	−30

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	Before	After	Change
Buyers pay	$100	$130	+30
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The law taxed sellers.

Buyers carry $30 of the $40, sellers $10.

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Tax revenue $= \$40 \times 270 = \$10{,}800$

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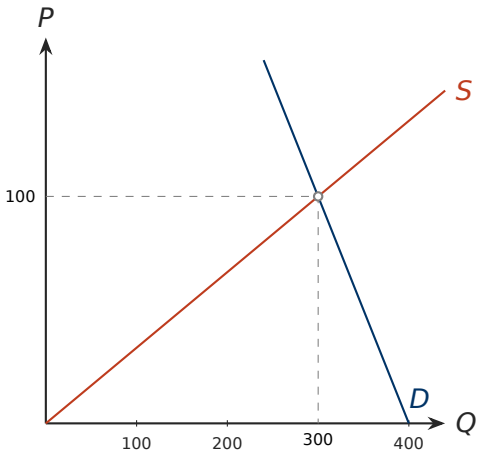
Three results

Buyers pay more, sellers keep less: both share the burden

The market shrinks: 30 trades that no longer happen

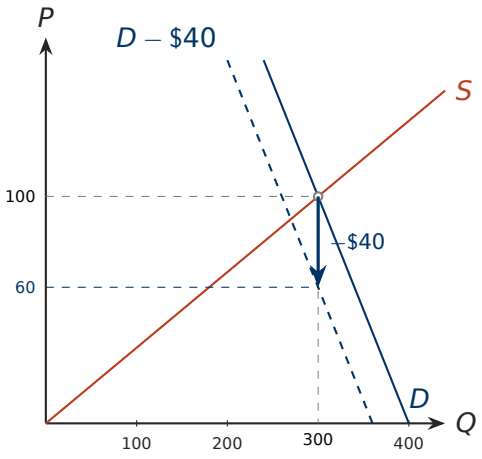
The price *wedge* between buyers and sellers equals the tax

Now tax the buyers instead



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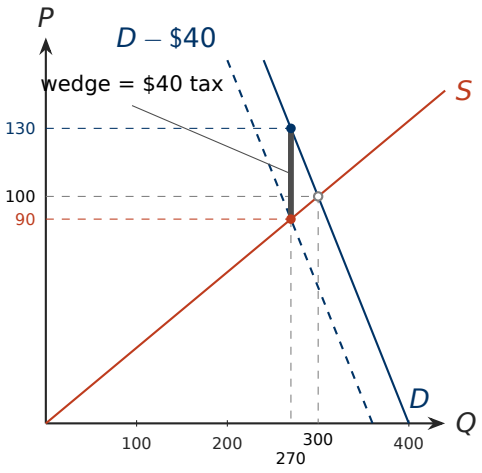
Before: 300 units at $100

Now they pay sellers only $60

D moves down $40:

$Q_d = 400 - (P + 40)$

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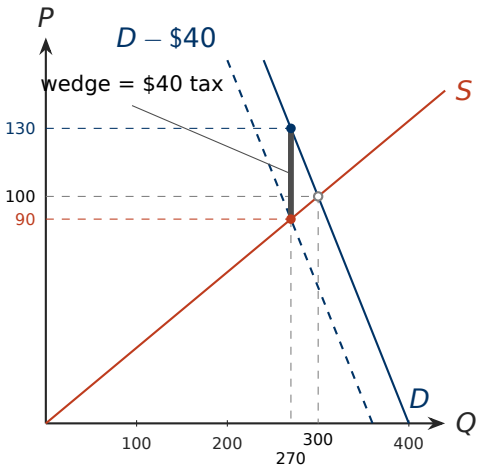
$Q_d = 400 - (P + 40)$

$360 - P = 3P \Rightarrow P = 90$

Sellers keep **90**, buyers pay **130**,

$Q = 270$

Now tax the buyers instead



Buyers owe \$40 on every unit bought

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Same answer

It does not matter which side the law taxes

Can Congress split the payroll tax?

Social Security and Medicare: 15.3% of wages

By law: 7.65% from the worker's paycheck, 7.65% from the employer

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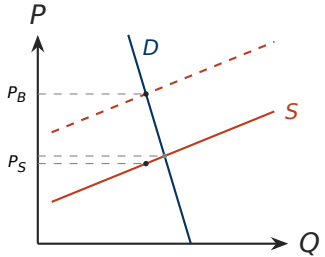
No law can decide that. The employer's share is part of what a worker costs; it comes out of wages.

Labor supply is fairly inelastic — most people work whatever the wage.

So workers carry most of the whole 15.3%.

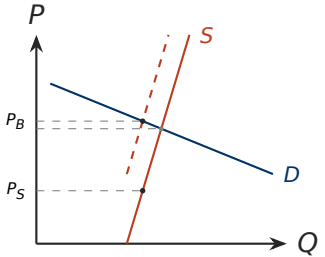
Who bears the tax: elasticity decides

Elastic *S*, inelastic *D*



buyers bear most

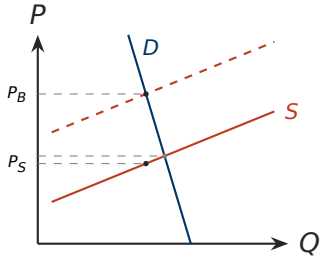
Inelastic *S*, elastic *D*



sellers bear most

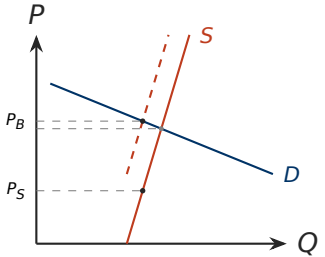
Who bears the tax: elasticity decides

Elastic *S*, inelastic *D*



buyers bear most

Inelastic *S*, elastic *D*



sellers bear most

The burden falls on the side that cannot walk away — the less elastic side.

Back to our $40 tax

At $P = 100$, $Q = 300$:

Demand: Q changes 1 per \$1 $\Rightarrow$
$E_d \approx \frac{1 \times 100}{300} = 1/3$

Supply: Q changes 3 per \$1 $\Rightarrow$
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Rule of thumb

Buyers' share $\approx \dfrac{E_s}{E_s + E_d}$

Here $\dfrac{1}{1 + 1/3} = \dfrac{3}{4}$ of \$40 = \$30

You will not need the formula on the exam. You will need the direction: the less elastic side pays more.

Exam-style: Fenway Park

Question

Fenway seats 25,000. Demand for tickets (thousands): $Q_d = 100 - 1.25P$.

(a) Find the equilibrium price.

(b) Boston adds a \$10 tax per ticket, paid by the buyer. Tickets sold? Fans' total price? Red Sox price? City revenue per game?

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(c) All on the Red Sox: vertical supply passes none of it on.

Application: the 1990 luxury tax

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Aim: make the rich pay

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Burden on boatbuilders and their workers, not buyers. Most of the tax was repealed in 1993.

Summary

Price controls

- Ceiling = legal maximum; floor = legal minimum
- Binding only if the ceiling is *below* or the floor *above* equilibrium
- Binding ceiling $\Rightarrow$ shortage, non-price rationing; binding floor $\Rightarrow$ surplus
- Fewer units trade either way
- Rent control: shortages grow in the long run. Minimum wage: pay up for some, jobs lost for others
- Vouchers and the EITC help without fixing the price

Taxes

- A tax puts a wedge between what buyers pay and sellers keep; the market shrinks
- Tax on sellers: S up by the tax. Tax on buyers: D down by the tax. Same outcome
- The law cannot decide who pays
- The less elastic side bears more of the burden

Next: review, Mon Oct 5.

Midterm 1: Wed Oct 7, Ch 1–5.